to consult Dennett's own account when it is published, rather than rely on my doubtless imperfect and impressionistic—maybe even embellished—one.

The psychologist Nicholas Humphrey, too, has developed a tempting hypothesis of how the evolution of a capacity to simulate might have led to consciousness. In his book, *The Inner Eye*, Humphrey makes a convincing case that highly social animals like us and chimpanzees have to become expert psychologists. Brains have to juggle with, and simulate, many aspects of the world. But most aspects of the world are pretty simple in comparison to brains themselves. A social animal lives in a world of others, a world of potential mates, rivals, partners, and enemies. To survive and prosper in such a world, you have to become good at predicting what these other individuals are going to do next. Predicting what is going to happen in the inanimate world is a piece of cake compared with predicting what is going to happen in the social world. Academic psychologists, working scientifically, aren't really very good at predicting human behaviour. Social companions, using minute movements of the facial muscles and other subtle cues, are often astonishingly good at reading minds and second-guessing behaviour. Humphrey believes that this 'natural psychological' skill has become highly evolved in social animals, almost like an extra eye or other complicated organ. The 'inner eye' is the evolved social-psychological organ, just as the outer eye is the visual organ.

So far, I find Humphrey's reasoning convincing. He goes on to argue that the inner eye works by self-inspection. Each animal looks inwards to its own feelings and emotions, as a means of understanding the feelings and emotions of others. The psychological organ works by self-inspection. I am not so sure whether I agree that this helps us to understand consciousness, but Humphrey is a graceful writer and his book is persuasive.

p. 78 A gene for altruistic behaviour …

People sometimes get all upset about genes 'for' altruism, or other apparently complicated behaviour. They think (wrongly) that in some sense the complexity of the behaviour must be contained within the gene. How can there be a single gene for altruism, they ask, when all that a gene does is encode one protein chain? But to speak of a gene 'for' something only ever means that a *change* in the gene causes a *change* in the something. A single genetic *difference*, by changing some detail of the molecules in cells, causes a *difference* in the already complex embryonic processes, and hence in, say, behaviour.

For instance, a mutant gene in birds 'for' brotherly altruism will certainly not be solely responsible for an entirely new complicated behaviour pattern. Instead, it will alter some already existing, and probably already complicated, behaviour pattern. The most likely precursor in this case is parental behaviour. Birds routinely have the complicated nervous apparatus needed to feed and care for their own offspring. This has, in turn, been built up over many generations of slow, step-by-step evolution, from antecedents of its own. (Incidentally, sceptics about genes for fraternal care are often inconsistent: why aren't they just as sceptical about genes for equally complicated parental care?) The pre-existing behaviour pattern—parental care in this case—will be mediated by a convenient rule of thumb, such as 'Feed all squawking, gaping things in the nest.' The gene 'for feeding younger brothers and sisters' could work, then, by accelerating the age at which this rule of thumb matures in development. A fledgling bearing the fraternal gene as a new mutation will simply activate its 'parental' rule of thumb a little earlier than a normal bird. It will treat the squawking, gaping things in its parents' nest—its younger brothers and sisters—as if they were squawking, gaping things in its own nest—its children. Far from being a brand new, complicated behavioural innovation, 'fraternal behaviour' would originally arise as a slight variant in the developmental timing of already-existing behaviour. As so often, fallacies arise when we forget the essential gradualism of evolution, the fact that adaptive evolution proceeds by small, step-by-step alterations of pre-existing structures or behaviour.